[W2] AI Mastery for Productivity & Innovation

Learning Outcomes (reference, from proposal):

- Develop fundamental understanding of **how AI works** to self-learn any AI tools
- Cultivate **mindset** to embrace AI to transform ways of working in the AI Era
- Explore most common **AI tools** and identify **practical use cases of AI** in daily work through demo & AI exploration
- Master the **six components of effective prompting** to get desired output from AI
- Activate **critical & creative thinking** before-, during- and after- prompting to ensure input/output relevance, quality and originality

Module mapping to agenda slots: **Slot 1 (11:30-1:00)** = Module 1 (55-min) + Module 2 (35-min) · **Slot 2 (2:00-3:30)** = Module 3 · **Slot 3 (4:00-5:30)** = Module 4

Note: prompting fundamentals (six components) are woven into Module 2 exploration and Module 4 project scaffolds rather than a standalone module, given the 1-day format. The AI Toolkit Exploration guide and Synergy Project workbook carry the prompting scaffolds.

Module 1 (55-min): Future of Work = Human x AI

Hook 5-min	AI is all around us — quick-fire spotting: · Rapid round: "Shout out one place AI already touched your life before 10am today" (Face ID, maps routing, spam filter, feed ranking, e-wallet fraud check) · Key message: **AI & digital is here to stay (whether we like it or not)** — the question is not if, but how we use it
CK 10-min	**Weak vs Strong AI** [draw on flipchart]: · Weak (narrow) AI — **SLIM**: **S**pecific task, **L**imited autonomy, **I**nstruction-based, **M**imic human intelligence ("It may seem smart, but it doesn't truly 'understand' like a human — just following patterns") · Strong AI — **FARR**: **F**ull autonomy, **A**daptive (learn, evolve, apply knowledge across different tasks), **R**emember & **R**eason (genuinely "understand" and make judgments) · The AI in sci-fi movies (Skynet) is Strong AI — **it does not exist yet**. Everything we use today is Weak AI.
CK 15-min	**The 4 Stages of AI Development** (latest version): · **1. Reactive Machines** [Weak AI]: follows pre-programmed rules, no memory, cannot learn. Examples: vending machines, spam filters, traffic light controllers. · **2. Limited Memory Machines** [Weak AI — where we are now]: short-term/selective memory, learns from patterns in stored data. Examples: Tesla's working memory ≈ 5-10 seconds; latest GenAI models hold up to ~1 million tokens (~750,000 words — like reading 10 novels simultaneously); reasoning models use chain-of-thought (pause → work step-by-step → respond); frontier models now run for many

	hours autonomously completing multi-step tasks with hundreds of tool calls. · "We are still at Stage 2, but advancing very fast. A bicycle and a Formula 1 car are both 'wheeled vehicles' — technically the same category. The gap? Enormous." · **3. Theory of Mind** [Strong AI — theoretical]: ToM is from developmental psychology (starts ~18 months in children). Current GenAI shows ToM-**like** behaviour (it passed children's ToM tests — but likely because the tests were in its training data). Metaphor: "Like someone who memorised a speech in French but can't hold a conversation in French. The surface looks right; the depth isn't there." · **4. Self-Aware AI** [hypothetical]: consciousness, sense of self, general intelligence on par with humans. Does not exist. The **Hard Problem of Consciousness**: you can't verify consciousness even in humans from behaviour alone. Stage jumps are dimensional, not linear. · Conclusion: AI lacks real-world understanding, creativity & critical thinking — but is superb at computing, efficiency & accuracy. **AI is a tool, not magic** — knowing its strengths and limits is the key.
Quiz (Gamified) 5-min	EdTech quiz to consolidate: 5-6 items on Weak/Strong AI and the 4 stages (include 1-2 deliberately tricky items, e.g. "ChatGPT understands your feelings — true/false?")
CK 10-min	**Human Intelligence: what humans (still) do better** — metacognition & the 4 Dimensions [handout: "4 Dimensions: What Humans Can Do Better Than AI"]: · The human edge: self-awareness, consciousness, emotions, and **metacognition** (thinking about thinking) · **1. Embodied Cognition** — we think with a body: felt experience, phenomenal consciousness, mind-body connection · **2. Adaptive Reasoning** — intuition + metacognition; causal understanding (why), not just correlational patterns (what usually follows what) · **3. Relational Intelligence** — affective empathy (feeling with), not just cognitive empathy (modelling); trust built between people · **4. Purposeful Agency** — meaning-making, intrinsic motivation, and accountability: humans can own a decision; AI cannot
Activity 7-min	Insight reflection (EdTech word clouds or sticky notes on two boards): · "What can AI do better than humans?" / "What can humans do better than AI?" · Debrief into the core concept: **Human-AI Synergy is the new way of working** — neither human-only nor AI-only
Case Study 3-min	Rapid case framing — AI's role vs human's role in 3 work areas (decision making, strategic planning, communication): · e.g. Decision making: AI = data-driven insights, predictions, consistency; Human = real-world context, ethics, judgement, the final call

	• Bridge: "So where exactly does AI plug into YOUR work? That's the toolkit — next."

Module 2 (35-min): AI Toolkit — Transform Ways of Working

CK 5-min	The **5 Areas of Work** where AI applies for productivity & innovation (walk-through, one vivid example each): • 1. **Planning & thinking** — AI as thought partner: brainstorm, recommend strategies, give feedback, ask you questions • 2. **Creating drafts of work** — first drafts of documents, decks, images, video, audio • 3. **Communication** — meeting minutes, translation, drafting & improving messages • 4. **Managing processes** [NEW] — automation workflows & AI agents that work FOR you • 5. **On-the-job learning** — explainers, deep research, source-based study (NotebookLM) Hand out the **AI Toolkit Exploration guide**; today's deep-dive is Area 4 — the newest superpower
CK + Demo (NEW) 20-min	**Managing Processes: Automation Workflows & Agent Building for Non-Technical People** **CK (8-min) — from using AI to delegating to AI:** • So far, AI helps when you show up and prompt it. **Managing processes = AI that works without you prompting each time.** • **Automation workflow** = Trigger → Steps → Action ("WHEN a form response arrives → summarise it → post to the team channel"). Deterministic: same steps, every time. Best for repetitive + predictable. • **AI agent** = goal + instructions + tools + knowledge. You define the outcome and guardrails; the agent decides the steps and handles variation. Best for tasks needing judgement within bounds. • Anatomy of an agent: model (the brain), instructions (the job description), tools/data it can use (the hands), and the boundaries you set • Rule of thumb: **if you can write the steps as a recipe → workflow; if you'd brief a junior colleague instead → agent** • Platform options — all no-code; choose what fits RCX's stack: **GenAI-native** (Custom GPTs, Gemini Gems, Claude Projects), **automation platforms** (Zapier, Make), **Microsoft ecosystem** (Power Automate, Copilot Studio), Lark(?), **open-source** (n8n) **Demo 1 (6-min) — build an automation live:** • Template-based: "When an email with attachment arrives → save to Drive/OneDrive folder → notify me in Teams/Slack" (Zapier or Power Automate) • Narrate the anatomy out loud: "Here's my TRIGGER… here are the STEPS… here's the ACTION." Show it firing with a real email.

	Demo 2 (6-min) — build an agent live: · Create a "Meeting Minutes Agent" (Custom GPT / Gem / Claude Project): paste role instructions + the team's minutes template, then feed it messy raw notes → structured minutes with action items · Key message: **"I built this in 5 minutes, with zero code, while you watched. You don't need IT's queue to start — you need a use case and oversight"** (oversight: this afternoon)
Activity 10-min	Guided AI toolkit exploration — **Try It Out!** (separate exploration guide) · Each participant tries ONE example: copy-paste an agent instruction set, or open an automation template and walk through its trigger → action · Pairs allowed (one screen, two brains); facilitators circulate to unblock logins/setup · Closing prompt (30 sec): "Note the ONE process of yours this could touch — you'll need it at 4pm." (feeds Module 4 project) *NDLM: curiosity converted into safe first hands-on success within minutes of the demo.*

Module 3 (90-min): Re-designing Work — Human-AI Synergy

CK 5-min	**A job is a bundle of tasks**: · "A job isn't one thing — it's many tasks. Some can be automated, some augmented with AI, some require human expertise." Example walk-through: Marketing Manager (data analysis → automate; content creation → augment; brand strategy → human) · WEF data: fastest growing & declining jobs — the common pattern: **repetitive, predictable tasks are being automated; human skills are the competitive edge** · The hard truth: "AI will not replace humans, but it **will** replace humans who don't use AI." So the question shifts: from "Will AI take my job?" to **"Which parts of my job can AI enhance, and where should I focus my human skills?"**
CK 10-min	The **AAH Matrix** — Automate, Augment, Human [handout]: · **Automate**: repetitive + predictable + low judgement → hand the whole task to AI/automation (with oversight) · **Augment**: judgement or creativity needed, but AI can draft/analyse/accelerate → human + AI together · **Human**: empathy, accountability, relationships, ethics, high-stakes judgement → human leads, AI stays out (or far in the background) · "Opportunities for augmenting humans are far greater than opportunities to automate existing tasks" — augmentation is where most value lives · When sorting, think about the **value** you want from the task, not just whether AI "can" do it

Quiz (Gamified) 10-min	Gamified AAH sorting quiz (EdTech): 10 RCX-flavoured tasks; teams sort each into A / A / H · Debrief the contested ones — disagreement is the learning moment (criteria over instinct)
Activity (Guided) 20-min	**Re-design your workflow** (worksheet or laptop): · Step 1 (5 min): list your recurring weekly responsibilities; pick ONE meaty process · Step 2 (10 min): break it into 4-8 activities, start-to-end workflow · Step 3 (10 min): sort each activity with the AAH Matrix · Step 4 (10 min): identify suitable tools per activity — including **automation/agent candidates** for anything marked Automate (use the 5-areas toolkit + exploration guide from Module 2) · Output: a before/after map of their role — visualising how AI transforms their work positively *Facilitators circulate; push back on "Human" labels driven by habit rather than value, and "Automate" labels on judgement-heavy tasks.*
Debrief 10-min	· Show full AAH example table; collect 2-3 share-outs of the most surprising re-design · Flag forward: **"Circle 1-2 activities you marked Automate. After the break, you will actually build one."** (feeds Module 4 project)
CK (35-mins)	How to write Instruction for automation & agent building

Module 4 (90-min): Human-AI Synergy Project

CK + Case Study 7-min	**The 3 States of Using AI** + over-reliance case study: · Case: "Alex", a busy junior associate, builds a client proposal entirely from one lazy prompt — generic copy, unchecked claims, irrelevant image, no thought on quality control. Debrief: What were the signs? What are the consequences? · The **3 states**: Underuse (your job gets replaced by AI — or by people using AI) / Over-rely (work quality drops, thinking skills & motivation degrade — the productivity paradox) / **Human-AI Synergy** (more strategic and innovative; quality AND productivity rise) · Flash the **9 Signs of Over-relying on AI**; quick self-reflection: "Which sign is closest to home?" (private, then optional shares)
CK 8-min	**The Human-AI Synergy method — thinking before, during, after** (the Burger): · **Before** [bun — human]: What's the objective/outcome? Can AI help? What do I want out of it? — the better your thinking, the better the input

	· **During** [patty — human-AI]: Am I satisfied? How do I re-prompt/iterate? Could another tool do better? — the better the input, the better the output · **After** [bun — human]: What do I edit/refine? What needs fact-checking? Is it authentic — where's my human touch? — the better the output, the better the outcome · Extension to today's build: when you **delegate a process** to a workflow/agent, the Burger doesn't disappear — it becomes the **design of the system**: Before = design & oversight, During = build & test, After = verify & own
CK 15-min	**Building Human Oversight into Automation & Agentic Workflows** · Why it matters: automation amplifies whatever you feed it — quality at scale, or **errors at scale, silently**. An unsupervised workflow doesn't get tired; it also doesn't notice it's wrong. · The 3 oversight models: ○ **Human-in-the-Loop (HITL)**: the system prepares, a human approves BEFORE the action executes (e.g. agent drafts the reply, you press send). Highest control, lowest speed. ○ **Human-on-the-Loop (HOTL)**: the system acts autonomously; a human monitors and can intervene (e.g. auto-filing with an audit log and exception alerts). Medium control, high speed. ○ **Human-out-of-the-Loop (HOOTL)**: full autonomy, review only after the fact. ONLY for low-stakes, reversible, well-tested processes. · Choosing the level: **match oversight to risk = stakes × reversibility**. High-stakes or irreversible (money out, external comms, personnel decisions) → always HITL. Low-stakes and reversible (internal filing, draft generation) → can graduate toward HOOTL. · Design principles: **approval gates** at consequential steps; an **escalation rule** ("if unsure or input is unusual → stop and ask me" — give the agent an escape hatch); **audit trail** (log what it did); a **kill switch** (know how to pause it); **least privilege** (only the access it needs); **earn autonomy** — start every new automation HITL and loosen oversight only with a track record, exactly like onboarding a new team member. · Non-negotiable: **accountability stays human. You own the output of your automation.**
Activity (Project) 60-min	**Human-AI Synergy Project** — each individual builds an automation workflow OR a simple agent for their OWN chosen use case (use the activity marked Automate in Module 3) · Guided by the **Human-AI Synergy Project workbook** (separate document) — strictly timeboxed: ○ Part 1 — Before You Build (15 min): choose use case, map the process, sort AAH, choose workflow vs agent, design oversight (HITL/HOTL) ○ Part 2 — Build & Test (25 min): write agent instructions or configure a workflow template; test with a real example; at least one iteration round

	◦ Part 3 — After: Verify & Own It (15 min): failure-mode check, oversight confirmation, human touch, rollout plan ◦ Buffer (5 min) · Individual submission: completed workbook + working prototype (or documented design if tool access blocked) · Facilitators circulate; triage rule: login/tool blockers → switch participant to the GenAI-native path (Custom GPT/Gem/Claude Project), which needs no extra accounts *NDLM: full cycle of error & feedback in a real build, consolidating the entire day's learning into a take-home artefact they will actually use at work.*

Wrap up Day 2 (30-min)

Share-out 15-min	· 3-4 volunteers demo their build (2-3 min each): the use case, workflow or agent, and **where the human stays in the loop** · Trainer highlights one strong oversight design per share
Reflection 10-min	· EdTech pulse check + reflection: "One process I will never do fully manually again" / "One thing I will NOT hand to AI — and why"
Preview Day 3 5-min	· "Day 1: yourself. Day 2: your work. Day 3: each other and this organisation — the most human day of all." Logistics reminder.